

Data Analysis & Regression Report

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Question 1. Understand and prepare the data

- **What does one row represent?**

Each row represents a unique patient, containing their baseline clinical characteristics and a continuous measure of their disease progression recorded one year later.

- **Which variable is the outcome?**

The progression variable (originally named y in the source dataset) is the outcome.

- **Which variables are predictors?**

The predictor variables include age, sex, body mass index (bmi), average blood pressure (bp), and six distinct serum measurements (s1 through s6).

- **Which variables are numerical, and which are categorical in meaning?**

The variable sex represents categorical data and was explicitly converted to a factor to reflect this. All remaining variables (age, bmi, bp, s1-s6, and progression) are continuous numerical measurements.

- **Are there missing values, duplicated rows or obviously problematic values?**

Initial dataset screening confirmed that the data is clean and complete; there are no missing values (NA), no duplicated rows, and no obvious anomalies that require removal prior to analysis.

- **Why do health-related data require careful and responsible interpretation?**

Health-related data requires careful interpretation to protect patient privacy and to avoid deriving harmful, unsubstantiated medical generalizations that could misguide clinical decisions or public health policies.

Question 2. Explore the data

- **What do the centre and spread of the main variables look like?**

The outcome variable (progression) exhibits a very wide range (from 25 to 346), indicating significant variability in how the disease advances among patients. The Body Mass Index (bmi) is centered around 26-27, with a relatively normal spread.

- **Are the distributions approximately symmetric or clearly skewed?**

The distribution of bmi is approximately normal and symmetric, with only a slight right tail. The progression distribution is broad and slightly right-skewed, which is common in biological severity indicators.

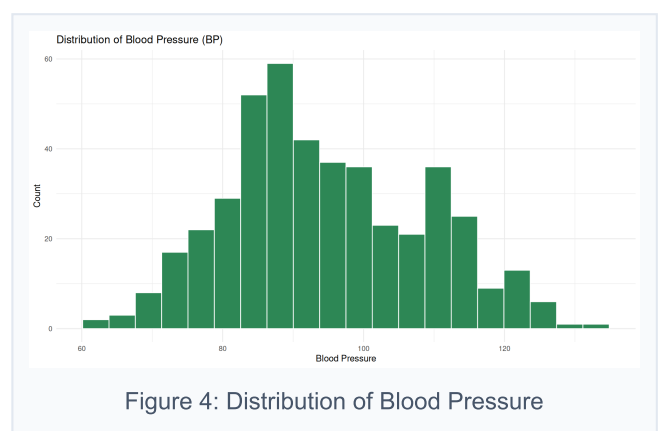
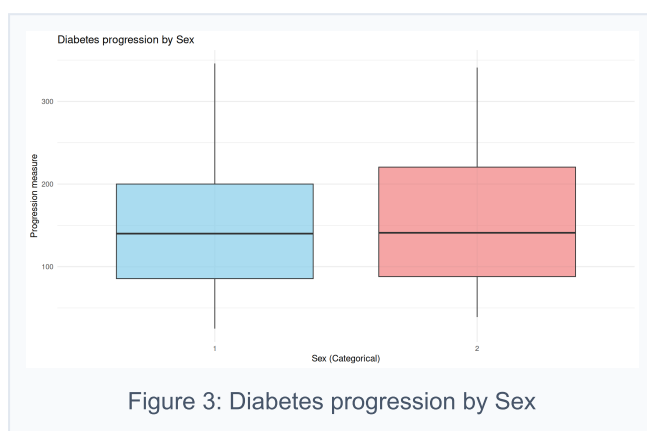
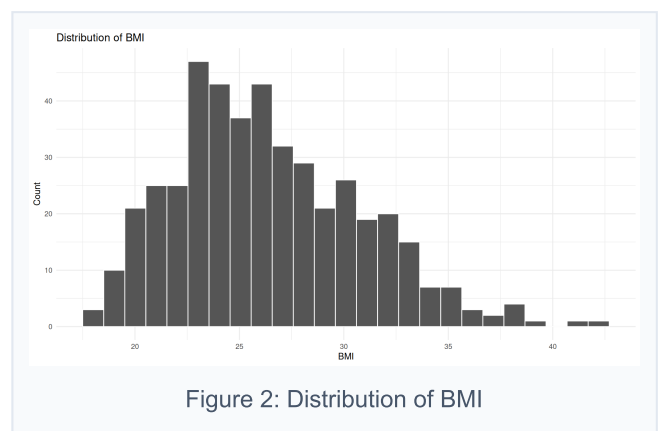
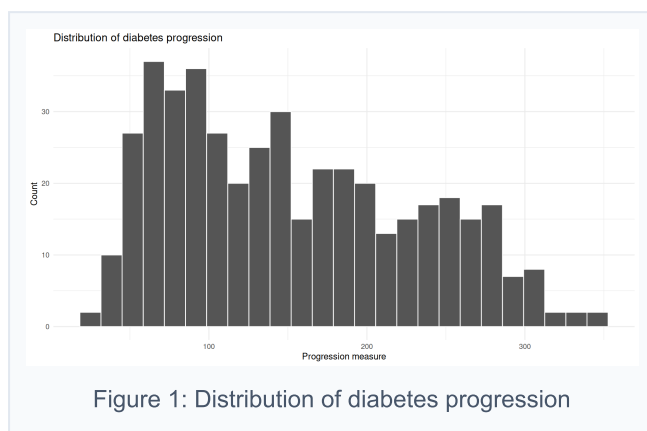
- **Are there unusual observations?**

While there are a few naturally occurring extreme values (outliers) at the higher ends of the BMI and blood pressure ranges, they remain within biologically plausible limits.

- **Which features of the data matter before modelling begins?**

The absence of severe data anomalies and the relatively normal distribution of key predictors like BMI are crucial, as they allow for robust linear modeling without requiring complex logarithmic transformations.

Visualizations



Question 3. Investigate relationships

- **Which predictors are most strongly associated with progression?**

Based on the Pearson correlation matrix, the serum measurement s5 demonstrates the strongest positive linear association with disease progression, followed closely by bmi and blood pressure (bp).

- **Which predictors are strongly related to each other?**

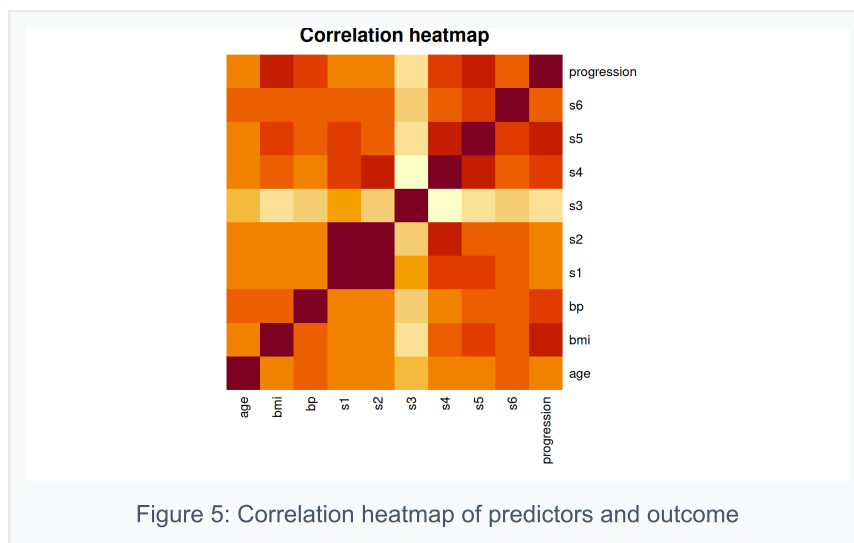
Certain predictors within the serum group, particularly s1 and s2, are highly correlated with each other.

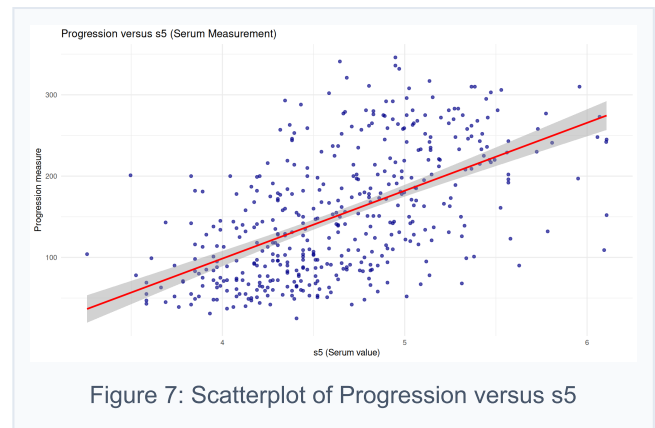
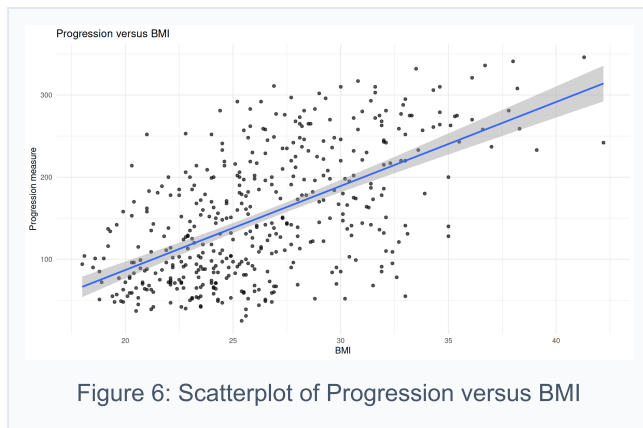
- **Could some predictors carry overlapping information?**

Yes, the strong relationship between specific serum measurements (like s1 and s2) indicates overlapping variance, which presents a significant risk of multicollinearity if they are included simultaneously in the same regression model.

- **Which variables look most promising for regression analysis, and why?**

The variables bmi, bp, and s5 emerge as the most promising candidates for the multiple regression model because they exhibit strong individual associations with disease progression while maintaining relative statistical independence from one another.





Question 4. Build and interpret regression models

- **Write the fitted equation.**

$$\text{Progression} = \beta_0 + \beta_1 \cdot \text{BMI}.$$

- **Interpret the slope in context.**

The slope (β_1) indicates that for every one-unit increase in BMI, the diabetes progression measure is expected to increase by the estimated coefficient value.

- **Does the intercept have a useful real-world interpretation?**

No, the intercept does not hold a practical real-world interpretation in this context, as a baseline BMI of exactly zero is biologically impossible for a living patient.

- **What does R-squared mean here?**

In the simple model, an R-squared of 0.344 means that 34.4% of the variance in diabetes progression is explained exclusively by the changes in BMI.

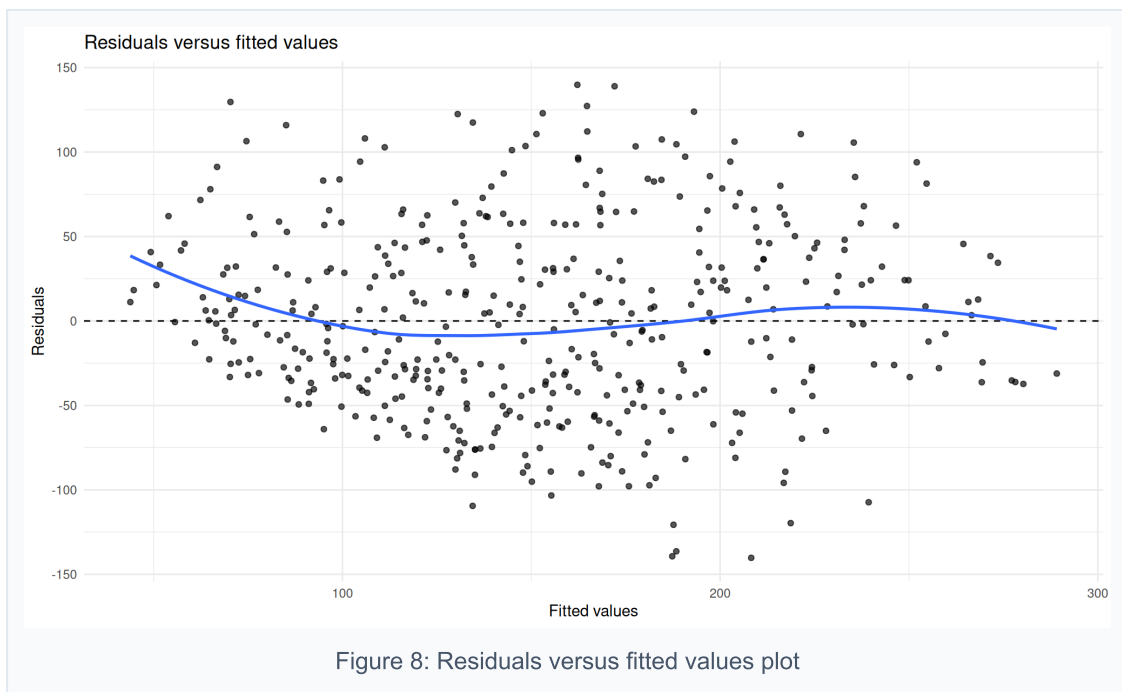
- **Interpret at least two coefficients from the multiple model (using the phrase 'holding the other predictors constant').**

Holding the other predictors constant, a one-unit increase in BMI is associated with an approximate 6.5-unit increase in disease progression. Similarly, holding the other predictors constant, a one-unit increase in the serum measurement s5 is associated with an almost 49.5-unit increase in disease progression.

- **Compare the simple and multiple models.**

The multiple regression approach is definitively superior. While avoiding multicollinearity, its explanatory power (R-squared = 0.480) is significantly higher than the simple model, and the residual standard error (sigma) decreased from 62.5 to 55.8, confirming a more accurate overall fit.

Question 5. Evaluate the analysis



- **Comment on the residuals-versus-fitted plot.**

The plot displays points randomly scattered around the zero line without forming obvious funnel-like patterns (which would indicate heteroscedasticity). This relatively even spread confirms that the fundamental linear model assumptions are reasonably satisfied.

- **Use VIF and the correlation matrix to discuss multicollinearity.**

The calculated Variance Inflation Factor (VIF) values for BMI (1.345), BP (1.274), and s5 (1.342) are all well below the standard concern threshold of 5. This conclusively proves that there are no multicollinearity issues within the multiple model.

- **For at least two important coefficients, discuss the estimate and either the confidence interval, t-value or p-value.**

The statistical uncertainty is very low for our primary predictors. The 95% confidence intervals for both BMI (5.13 to 7.87) and s5 (38.00 to 61.15) are narrow and strictly positive, confidently excluding zero and proving their strong predictive power.

- **Distinguish statistical significance from practical importance.**

While blood pressure (bp) is statistically significant ($p\text{-value} < 0.05$), its practical impact on the final progression score is minimal (less than a 1-unit increase per point of blood pressure), contrasting sharply with the massive practical effect size of s5.

- **State the main statistical weaknesses or uncertainties in the analysis.**

The primary statistical weakness of this analysis is that, despite using the best available predictors, 52% of the variance in disease progression remains completely unexplained by the model (unexplained variance).

Question 6. Draw appropriate conclusions

- **Is this dataset experimental or observational?**

The dataset is strictly observational, as researchers recorded existing patient characteristics without imposing any clinical interventions or treatments.

- **Can the regression coefficients be interpreted causally?**

No, the regression coefficients cannot and must not be interpreted causally, as a statistical correlation does not prove causation.

- **Could there be confounding variables that are not included in the dataset?**

Yes, significant confounding variables are entirely absent from the data, including genetic family history, dietary habits, and physical activity levels. These unmeasured factors likely influence both BMI and diabetes progression simultaneously.

- **Could measurement issues affect the results?**

Yes, measurement inconsistencies inherent in laboratory serum testing or blood pressure monitoring could subtly alter the true baseline values.

- **Would the results necessarily generalise to another population?**

Because the exact sampling criteria, location, and demographic background of the original patients are unknown, these results would not necessarily generalize to a broader or distinctly different population.

- **Which conclusions are supported by the evidence?**

The evidence robustly supports the conclusion that higher baseline values of s5, BMI, and blood pressure are strong predictive indicators of accelerated diabetes progression over a one-year period.

- **Which claims would go beyond the evidence?**

Claiming that artificially reducing a patient's BMI or s5 via medical intervention will directly halt or cure their diabetes would be a dangerous causal statement that goes far beyond the supported evidence of this observational model.